

Q.1

Y

- a) Y is even ~~is~~
 Y is even when its 2 or 6 or 0

$$\therefore \text{probability} = \frac{1}{27} + \frac{1}{9} + \frac{1}{27} + \frac{1}{9} + \frac{1}{9} + \frac{1}{9} + \frac{4}{27}$$

$$= \frac{1+3+1+3+3+3+4}{27} = \frac{18}{27} = \frac{1}{3}$$

- b) XY is odd
 multiplication is odd when both numbers are odd. It's when $(X, Y) = \{(1, -1), (3, -1)\}$

$$\therefore \text{probability} = \frac{2}{9} + 0 = \frac{2}{9}$$

- c) $X > 0$ $Y \geq 0$
~~(X, Y) = (1, 3)~~ when $X = 1, 3$
 and $Y = 0, 2, 6$

$$\therefore \text{probability} = \cancel{\frac{1}{27}} \left(0 + \frac{1}{9} + \frac{1}{9} \right) + \left(0 + \frac{1}{9} + \frac{4}{27} \right)$$

$$= \frac{2}{9} + \frac{3}{27} + \frac{4}{27}$$

$$= \frac{6+3+4}{27} = \frac{13}{27}$$

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9.2

$$f(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{ow} \end{cases}$$

$$F_{X/Y}(t) = P\{X/Y \leq t\} \\ = P\{X \leq Yt\}$$

$$\therefore P\{X/Y \leq t\} = \int_{-\infty}^{\infty} \int_0^{yt} f(x) g(y) dx dy$$

$$F(t) = \int_{-\infty}^{\infty} \left(\frac{1 - e^{-yt}}{g(y)} \right) g(y) dy$$

$$F(t) = \int_{-\infty}^{\infty} (1 - e^{-yt}) g(y) dy$$

$$F(t) = \int_{-\infty}^{\infty} g(y) dy - \int_{-\infty}^{\infty} e^{-yt} g(y) dy$$

$$\therefore F(t) = 1 - M(-t)$$

□

Q.3. $X, Y \rightarrow$ iid random variables
with common density function:

$$f_X(x) = f_Y(y) = f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

To find PDF of $XY, X/Y, \min\{X, Y\}, \max\{X, Y\}$

a) The density function for product is
given by.

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{|x|} f_{XY}\left(x, \frac{z}{x}\right) dx$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y\left(\frac{z}{x}\right) \frac{1}{|x|} dx$$

Since $f_X(x) = 1$ if $0 \leq x \leq 1$ and 0 otherwise

$$f_Z(z) = \int_0^1 f_Y\left(\frac{z}{x}\right) \frac{1}{x} dx$$

Now, $f_Y\left(\frac{z}{x}\right) = 0$ unless $0 \leq \frac{z}{x} \leq 1$

and then it is 1. $\left(0 \leq \frac{z}{x} \leq 1\right)$
 $x \leq 1.$

$$f_Z(z) = \int_z^1 (1) \frac{1}{x} dx = \boxed{-\ln(z)}$$

$$\therefore \text{PDF of } XY \text{ is } \begin{cases} -\ln(z) & 0 < z \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

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Q.3
(b)

The density function of ratio
is given by $z = x/y$

$$f_2(z) = \int_{-\infty}^{\infty} |y| f_x(zy) f_y(y) dy$$

$$= \int_0^{\infty} |y| f_x(zy) dy$$

$$0 \leq zy \leq 1$$

$$0 \leq y \leq \frac{1}{z}$$

For $z > 1$

$$f_2(z) = \int_0^{1/z} |y| dy = \boxed{\frac{1}{2z^2}}$$

$$(c) \text{ PDF} = \begin{cases} \frac{1}{2z^2} & z > 1 \\ 0 & \text{or} \end{cases}$$

(c) For non PDF is given by

~~$$f_2(z) = \int_{-\infty}^{\infty} |y| f_x(zy) f_y(y) dy$$

$$\text{PDF} = 2 \cdot f(x) \cdot \int_{-\infty}^x f(y) \cdot dy$$~~

Q.3 (c) The density function of $\min(X, Y)$ is given by

~~$$f_z(z) = \frac{d}{dz} F_z(z) = \frac{d}{dz} (1 - [1 - F(z)]^2)$$

$$= 2[1 - F(z)] F'(z) = 2f(z)[1 - F(z)]$$~~

~~(where $F_z(z) = 1 - P\{Z > z\}$)~~

$$f_z(z) = \frac{d}{dz} F_z(z) = \frac{d}{dz} (1 - [1 - F(z)]^2)$$

$$= 2[1 - F(z)] F'(z) = 2f(z)[1 - F(z)]$$

(where $F_z(z) = 1 - P\{Z > z\}$)

$$= 1 - [1 - F(z)] \cdot [1 - F(z)]$$

$$= 1 - [1 - F(z)]^2$$

$$P\{X > z\} = P\{Y > z\} = 1 - F(z) = \int_z^{\infty} f(x) dx$$

$$\therefore \text{PDF} = 2 \cdot f(z) \int_z^{\infty} f(x) dx \quad \text{for } 0 < z < \infty$$

$$= 2 \cdot 1 \cdot \int_z^{\infty} 1 \cdot dx$$

$$= 2(1 - z)$$

$$\therefore \text{PDF} = \begin{cases} 2(1 - z) & 0 < z < 1 \\ 0 & \text{otherwise} \end{cases}$$

Q.3 (d)

for $\max(X, Y)$

$$PDF = 2 \cdot f(z) \cdot \int_{-\infty}^z f(u) du \quad \text{for } 0 < z < 1$$

$$= 2 \cdot 1 \cdot \int_0^z 1 \cdot du$$

$$= 2z$$

$$\therefore PDF = \begin{cases} 2z & 0 < z < 1 \\ 0 & \text{otherwise} \end{cases}$$

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Q.4.

(a) Let X and Y be 2 RV with joint PDF $f(x, y)$

$$\therefore E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy$$

$$\therefore \text{we know } f_{Y/X}(y/x) = \frac{f(x, y)}{f_X(x)}$$

$$\therefore E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{Y/X}(y/x) f_X(x) dx dy$$

$$\Rightarrow E(XY) = \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} y f_{Y/X}(y/x) dy \right) f_X(x) dx$$

$$\therefore E(XY) = \int_{-\infty}^{\infty} x E[Y|X=x] f_X(x) dx$$

□

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Q.4) (b)

If $E[Y|X=x] = \mu$ is independent of X
then $\forall x, E[Y|X=x] = \mu$

$$E[E[Y|X]] = \int_{-\infty}^{\infty} \mu f_X(x) dx = \mu$$

$$\therefore E[Y] = \mu \quad (\because E[E[Y|X]] = E[Y])$$

Now,

$$\therefore \text{Var}(Y) = \text{Var}(E[Y|X]) + E[\text{Var}(Y|X)]$$

$$(\because \text{Var}(E[Y|X]) = \text{Var}(\mu) = 0 \text{ as } \mu = \text{constant})$$

$$\Rightarrow \text{Var}(Y) = E[\text{Var}(Y|X)]$$

$$\therefore \text{Var}(Y) = \int_{-\infty}^{\infty} \text{Var}[Y|X=x] f_X(x) dx$$

□

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Q.5

(a) Tom $\rightarrow f(x) = \begin{cases} 5e^{-5x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

Harry $g(y) = \begin{cases} 3e^{-3y} & \text{if } y \geq 0 \\ 0 & \text{otherwise} \end{cases}$

(a) For any time t , for tom to solve before harry,

~~Let X and Y be~~

Time taken by tom < Time taken by harry

Let X = Time taken by tom to solve a q.
& Y = Time taken by harry to solve a q.

$f(x, y) = f(x) \cdot g(y)$ ($\because X$ and Y are independent RV)

$$f(x, y) = \begin{cases} 15e^{-5x-3y} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$P(Y \leq X) = \int \int_{y \leq x} f(x, y) dx dy$$

$$= \int_{x=0}^{\infty} \int_{y=0}^x (15e^{-5x-3y}) dy dx$$

$$= \int_{x=0}^{\infty} 5e^{-5x} (1 - e^{-3x}) dx$$

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g.s (a) ~~calcd~~

$$= \int_{n=0}^{\infty} 5e^{-5n} - 5e^{-3n} dn$$

$$= 5 \left[\frac{e^{-5n}}{-5} + \frac{e^{-3n}}{3} \right]_0^{\infty} = \frac{3}{8}$$

For tom to be faster than harry
we have to find $P(X < Y)$

$$P(X < Y) = 1 - \{P(X \geq Y)\}$$

$$= 1 - 3/8 = \boxed{5/8}$$

$$\therefore \boxed{P(X < Y) = \frac{5}{8}}$$

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q. 5 (b)

$$P(X < Y | X > 1) = \frac{P((X < Y) \cap (X > 1))}{P(X > 1)}$$

$$P((X < Y) \cap (X > 1)) = \int_1^{\infty} \left(\int_x^{\infty} (g(y)) dy \right) f(x) dx$$

$$= \int_1^{\infty} 5e^{-8x} dx = \frac{5}{8} e^{-8}$$

$$P(X > 1) = \int_1^{\infty} 5e^{-5x} dx = e^{-5}$$

$$P(X < Y | X > 1) = \frac{\frac{5}{8} e^{-8}}{e^{-5}} = \frac{5}{8} e^{-3}$$

$$\therefore P(X < Y | X > 1) = \frac{5}{8} e^{-3}$$

9.5(c)

$$P(X-Y > 1) + P(Y-X > 1)$$

$$P(X-Y > 1) = \int_{x=1}^{\infty} \int_{y=0}^{y=x-1} 15 e^{-5x} e^{-3y} dy dx$$

$$= 15 \int_{x=1}^{\infty} e^{-5x} \int_{y=0}^{y=x-1} \frac{(-e^{-3y})}{3} dy dx$$

$$= 5 \int_{x=1}^{\infty} e^{-5x} (-e^{-3(x-1)} + 1) dx$$

$$= 5 \int_{x=1}^{\infty} e^{-5x} - e^{-x+3} dx$$

$$= 5 \left(\frac{e^{-5}}{5} - \frac{e^{-5}}{8} \right) = \boxed{\frac{3}{8} e^{-5}}$$

$$P(Y-X > 1) = 15 \int_{x=0}^{\infty} \int_{y=x+1}^{\infty} e^{-5x} e^{-3y} dy dx$$

$$= 5 \int_{x=0}^{\infty} e^{-3x} \int_{y=x+1}^{\infty} e^{-3y} dy dx$$

$$= 5 \int_{x=0}^{\infty} e^{-5x} \cdot e^{-3x-3} dx$$

$$= \boxed{\frac{5}{8} e^{-5}}$$

$$\therefore P = \frac{3}{8} e^{-5} + \frac{5}{8} e^{-5} = \boxed{e^{-5}}$$

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Q.6 Correlation between $X_1 - X_2$ and $X_2 + X_3$ is given by

$$\rho = \frac{\text{Cov}(X_1 - X_2, X_2 + X_3)}{\text{SD}(X_1 - X_2) \text{SD}(X_2 + X_3)}$$

$$\text{Cov}(X_1 - X_2, X_2 + X_3)$$

$$= \text{Cov}(X_1, X_2) + \text{Cov}(X_1, X_3) - \text{Var}(X_2) - \text{Cov}(X_2, X_3)$$

$$= 0 + 0 - \text{Var}(X_2) - \text{Cov}(X_2, X_3)$$

$$= -\sigma_2^2$$

($\because \text{Cov}(X_1, X_2) = \text{Cov}(X_1, X_3) = \text{Cov}(X_2, X_3) = 0$
as X_1, X_2 and X_3 are independent)

$$\text{SD}(X_1 - X_2) = \sqrt{\text{Var}(X_1 - X_2)} = \sqrt{\text{Var}(X_1) + \text{Var}(X_2) - 2\text{Cov}(X_1, X_2)}$$

$$= \sqrt{\sigma_1^2 + \sigma_2^2}$$

$$\text{SD}(X_2 + X_3) = \sqrt{\text{Var}(X_2 + X_3)} = \sqrt{\sigma_2^2 + \sigma_3^2}$$

$$\therefore \rho = \frac{-\sigma_2^2}{\sqrt{\sigma_1^2 + \sigma_2^2} \cdot \sqrt{\sigma_2^2 + \sigma_3^2}}$$

our reqd. correlation